# 2025 UBS Finance Challenge ~ Pair Trade Stock Pitch

**Region：** Hong Kong Stock Market

**Sector：** Industrial ( Automobile)

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# Summary of Rationale - Long Tesla (Tsla.US) Short BYD (1211.HK)

## ☐ Business and Financial Analysis

- Tesla should NOT be valued merely as an EV company but as a diversified technology and energy company with multiple high-growth business segments, undervalued by current market cap

## ☐ Quantitative Analysis

- BYD and Tesla's price spread is statistically robust and mean-reverting based on past 5-year stock price trend
- ARIMA model predicts that it is good timing to **Long Tesla short BYD** by mid-May 2025

## ☐ Trading Strategy and Risk Control

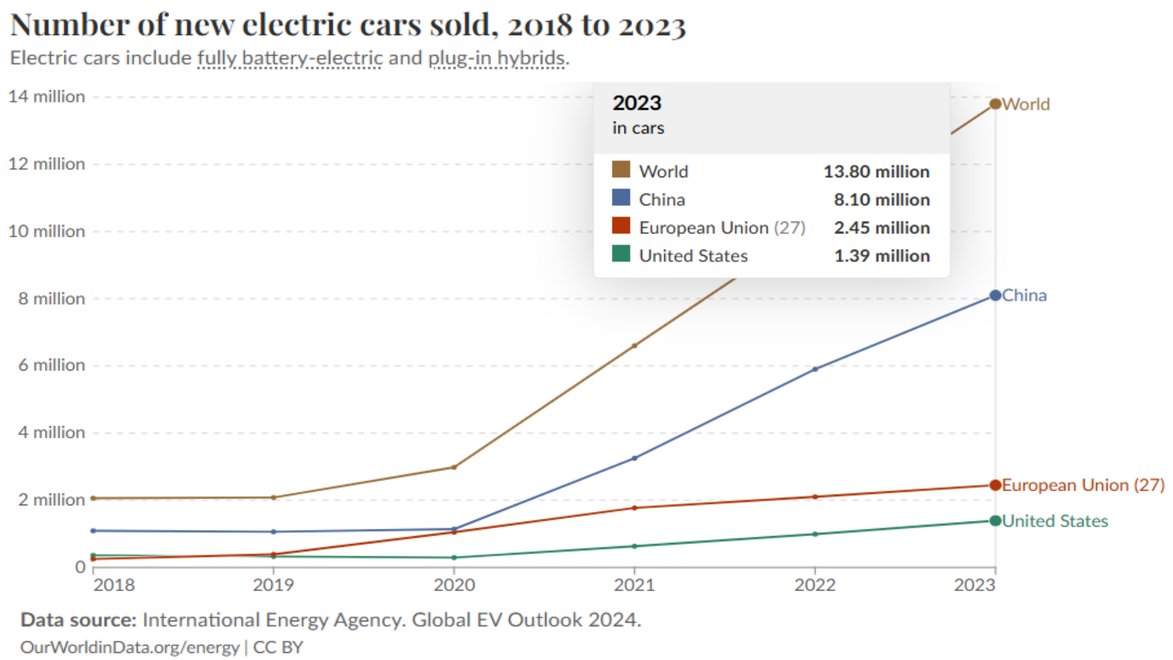
- Highlight trading strategies for **entry, exit and stop-loss timing**
- Summarize **key risks** for the pair stock trading

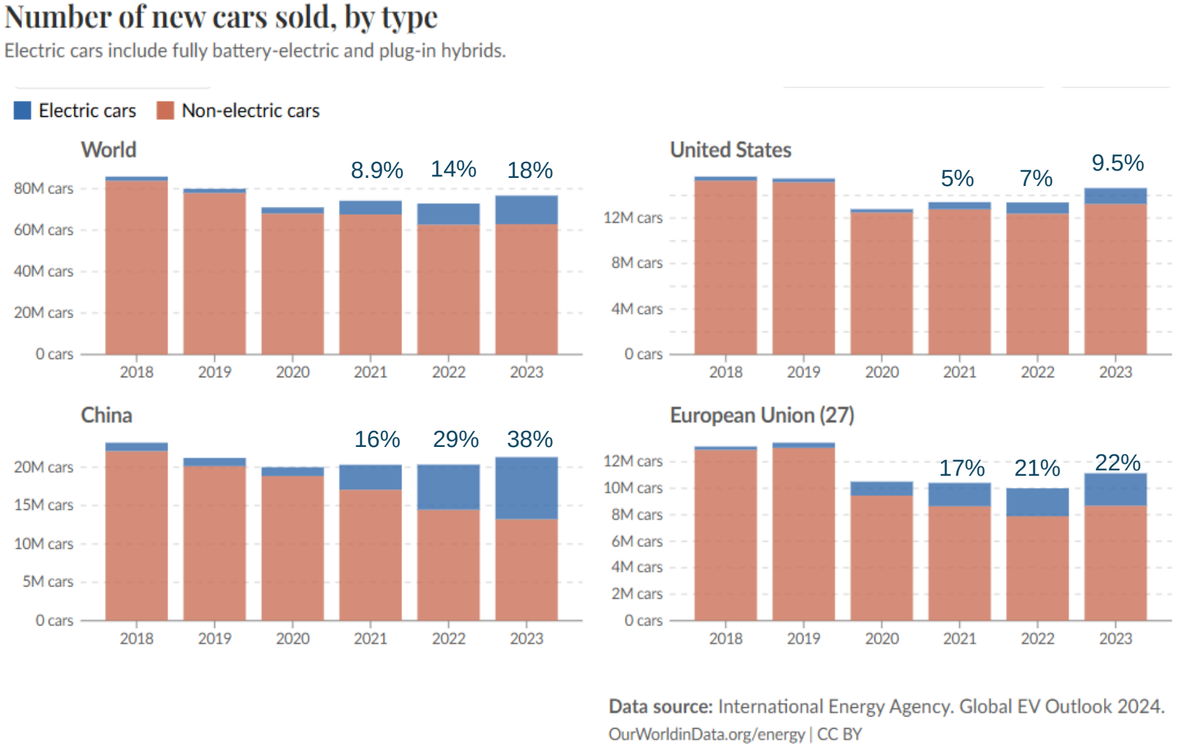



| | Apr 29, 2020 | Apr 30, 2025 | | 5-yr Growth % |
|---|---|---|---|---|
| Tesla Inc | $46.75 | $279.69 | +$232.94 | ↑ 486.42% |
| BYD Ord Shs H | $48.80 | $370.20 | +$321.40 | ↑ 658.61% |
| Nasdaq Composite | 8,604.95 | 17,099.98 | +8,495.03 | ↑ 97.72% |
| Hang Seng Index | 24,643.59 | 22,119.41 | -2,524.18 | ↓ 10.24% |

# Why Electric Vehicle (EV) Sector? - *Long Tesla (Tsla.US) Short BYD (1211.HK)*

- In 2024, the global electric vehicle (EV) market achieved **25%** growth to sales of **17 million** units (both BEVs and PHEVs) or **$380 billion** and expected to surpass **20 million** units in 2025, approximately **20%** penetration of global car sales
- Top two leaders: **BYD (22% of global market)** with volume-driven strategy **vs Tesla (17% of global market)** with platform economics
- EV sector has exhibited divergent price volatility in Hong Kong and NASDAQ market, reflecting distinct geopolitical, regulatory, and market structure dynamics. The 3-year average volatility of stock prices of EV indices in both markets is **36%** (HK) and **29%**(NASDAQ)



# Business and Financial Analysis (1) - *Long Tesla (Tsla.US) Short BYD (1211.HK)*

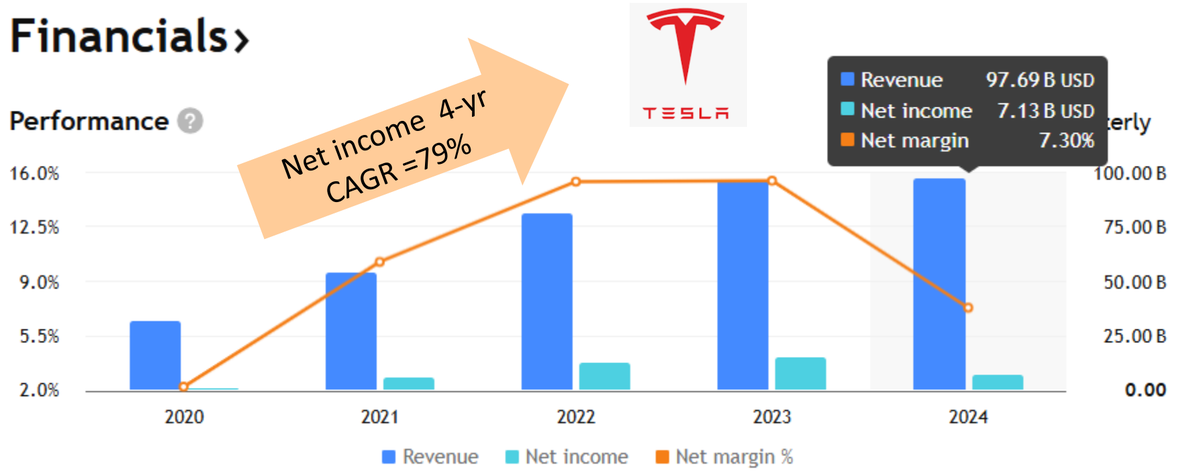
## Why Long Tesla?

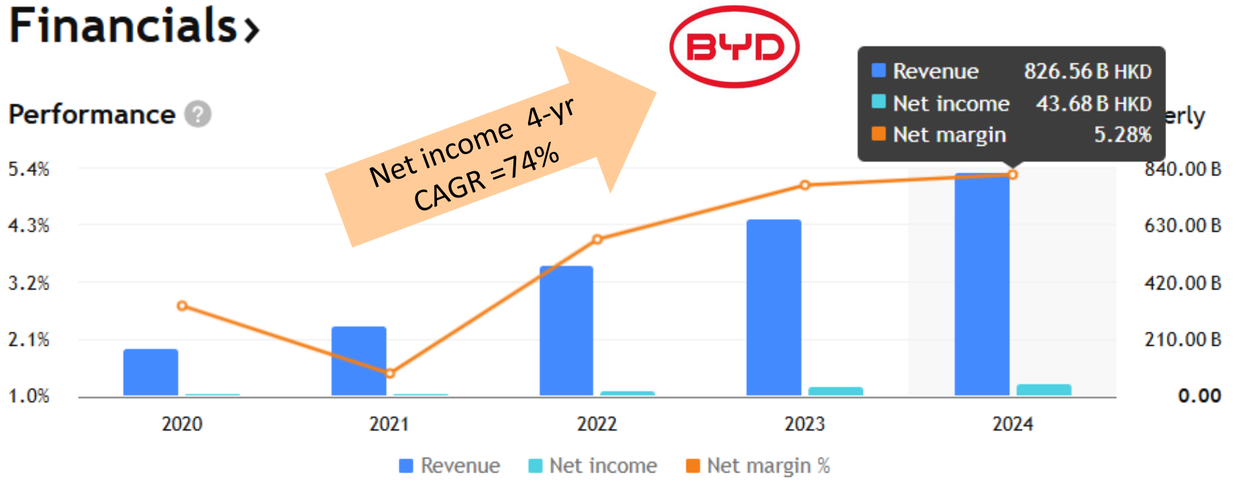
### ☐ Historical Performance

- **4-yr CAGR of 33%** in revenue and **79%** in net profit, implying more profitable and faster bottom line growth than that of BYD

### ☐ Valuation Adjustment Catalysts

- Full Self-Driving (FSD) Adoption will launch in Austin in June 2025, targeting $0.25/mile vs. Uber's $1.50 could capture 5% of the US ride-hailing market by 2026, adding **$4.2 billion in EBITDA**
- Tesla's energy storage **(10% of total revenue)** provides a hedge against automotive cyclicality for long-term upside
- Tesla's pivot to AI and robotics leverages its software expertise to unlock higher-margin revenue streams, justifying its premium valuation

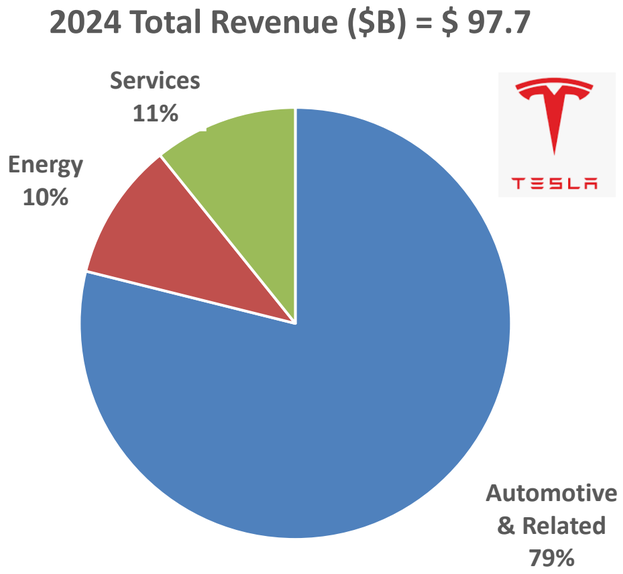


# Business and Financial Analysis (2) - *Long Tesla (Tsla.US) Short BYD (1211.HK)*

## ☐ Sum-of-Segment growth-adjusted P/E valuation unlock Tesla's intrinsic higher value

- Tesla trades at P/E **164x** at 2024 earnings, higher ratio compared to traditional automotive manufacturers, but **Sum-of-Segment analysis** reveals that the market may **undervalue** Tesla's potential in its non-automotive businesses, particularly in energy storage and autonomous driving technology
- By applying an automotive P/E multiple to the EV business and growth-adjusted P/E multiple to the energy and services segments, a higher valuation than Tesla's current market by **from 8.9% to 12.2%** can be justified given conservative assumptions and sensitivity analysis
- Key assumptions include the next six-year segment growth rate, net margin in 2030, P/E multiple in 2030 and three cases of discount rate (12%, 15% and 18%)




| Segment | 2024 Revenue | 2030 Revenue | 6-year CAGR % | 2030 Net Margin % | 2030 Net Profit | 2030 P/E (x) | 2030 Terminal Value | Present Value@12% | Present Value@15% | Present Value@18% |
|---|---|---|---|---|---|---|---|---|---|---|
| Automotive | $77.1 B | $103.3 B | 5% | 8% | $8.3 B | 12x | $99.1 B | | | |
| Energy | $10.1 B | $23.3 B | 15% | 12% | $2.8 B | 18x | $50.4 B | | | |
| Services/FSD | $10.5 B | $24.3 B | 15% | 15% | $3.6 B | 20x | $73.0 B | | | |
| **Total** | **$97.7 B** | **$150.9 B** | **7.5%** | **9.7%** | **$14.7 B** | **15.1x** | **$ 222.5 B** | **$ 112.7 B** | **$ 96.2 B** | **$ 82.1 B** |
| | | | | | | | **% Current Market Cap ($ 92.4 B)** | **12.2%** | **10.4%** | **8.9%** |

# Business and Financial Analysis (3) - *Long Tesla (Tsla.US) Short BYD (1211.HK)*

## Why Short BYD ?

### ☐ Historical Performance

- BYD's Free Cash Flow (FCF) margin declined from **10.3%** in 2022 to **6.4%** in 2024 despite higher volumes

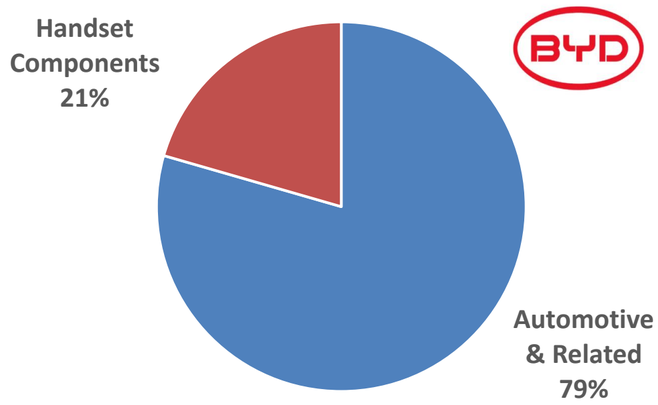
### ☐ Valuation Adjustment Catalysts

- BYD's current low P/E multiples reflect market skepticism about sustaining margins amid price wars. **10%** decline in average selling price (ASP) would erase 2025E net income growth despite volume gains
- BYD's lower R&D capitalization **(currently 2% vs. Tesla' 14%)**, eroding its value proposition and reliance on China government subsidies **(12% of 2024 revenue)** might not sustain in the future.
- Trade Barrier: EU imposed **17–36% tariffs** and the US hiking duties to **145%** will bring geopolitical disruptions to BYD's export-led growth
- BYD's local production is time-consumed and the competition in China market is more fierce

| Valuation Indicator | Tesla | BYD |
|---|---|---|
| 2025E Revenue Growth% | 12% | 30% |
| 2025E Net Margin | 8.2% | 5.1% |
| P/E(x) 2025E | 58x | 17x |
| EV/EBITDA (x) 2025E | 34x | 9x |
| R&D % revenue | 14% | 2% |
| 3-yr Cum. Capex / FCF (1) | 2.7x | 3.2x |
| 3-yr Cum. FCF / Delivery (1) | $5,872 | $3,961 |

*[1] Cumulative Free Cash Flow (FCF) for the years from 2022 to 2024*

**2024 Total Revenue ($B) = $ 107.1**



# Quantitative Analysis (1) - *Long Tesla (Tsla.US) Short BYD (1211.HK)*

## Correlation and Cointegration Test

| Metric | |
|---|---|
| **Hedge ratio** | |
| Hedge ratio | 0.897 |
| **CADF Test** | |
| Test Statistic | -3.273 |
| p-value | 5.85e-02 |
| Critical Value 1% | -3.902 |
| Critical Value 5% | -3.339 |
| Critical Value 10% | -3.047 |
| **ADF Test** | |
| Test Statistic | -3.270 |
| p-value | 1.63e-02 |
| Number of Lags Used | 10 |
| Number of Observations Used | 1833 |
| Critical Value 1% | -3.434 |
| Critical Value 5% | -2.863 |
| Critical Value 10% | -2.568 |
| **Spread SD** | |
| Spread SD | 0.227 |
| **Pearson r** | |
| Pearson r | 0.866 |
| **Num zero-crossings** | |
| Num zero-crossings | 75 |
| **Hurst Exponent** | |
| Hurst Exponent | 0.426 |
| **Half-life of mean reversion** | |
| Half-life of mean reversion | 51.6 |
| **% days within historical 2-SD band** | |
| % days within historical 2-SD band | 96.0% |

- Multiple statistics **(CADF and ADF)** proved the spread (price difference) between the two stocks reverts to a mean, confirming cointegration and enabling mean-reversion trading

### Mathematical Notes

### Statistics Implication

The hedge ratio ($\beta$) is computed using OLS regression on log prices of BYD ($X_t$) and Tesla ($Y_t$): $X_t = \alpha + \beta Y_t + \epsilon_t$

Here, $\beta = 0.897$ indicates that for a 1% change in Tesla's log price, BYD's log price changes by 0.897%.,The spread is then: $Z_t = X_t - \beta Y_t$

1. For two series $X_t$ (BYD) and $Y_t$ (Tesla), cointegration implies there exists a $\beta$ such that the spread $Z_t = X_t - \beta Y_t$ is stationary. The CADF test regresses $X_t$ on $Y_t$ to estimate $\beta$ (hedge ratio) and tests the stationarity of the residuals
2. . Applied to the spread $Z_t$, this tests for stationarity. A stationary spread (p-value < 0.05) suggests mean-reverting behavior, crucial for pairs trading. The test statistic is compared to critical values.

1. The test statistic ($-3.273$) is more negative than the 10% critical value ($-2.862$) but less negative than the 5% critical value ($-3.339$). The p-value (0.0585) is slightly above 0.05, suggesting that it's moderate evidence of cointegration between BYD and Tesla, meaning their prices have a stable long-term relationship, which is a prerequisite for pairs trading.
2. The ADF test on the spread (statistic = $-3.270$, p-value = 0.0163) indicates the spread is stationary at the 5% level (critical value = $-2.863$), confirming mean-reverting behavior suitable for pairs trading.

$$r = \frac{\sum(x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum(x_i - \bar{x})^2 \sum(y_i - \bar{y})^2}}$$

Pearson Correlation measures the linear relationship between the two stocks' log prices.

The high correlation (0.866) between BYD and Tesla's log prices indicates that the two stocks move together closely, which is expected for a cointegrated pair. This supports the rationale for pairs trading, as the spread captures deviations from this relationship.

The spread crosses its mean 75 times over the period, indicating frequent mean reversion. This is a positive sign for pairs trading, as it suggests the spread oscillates around the mean regularly.

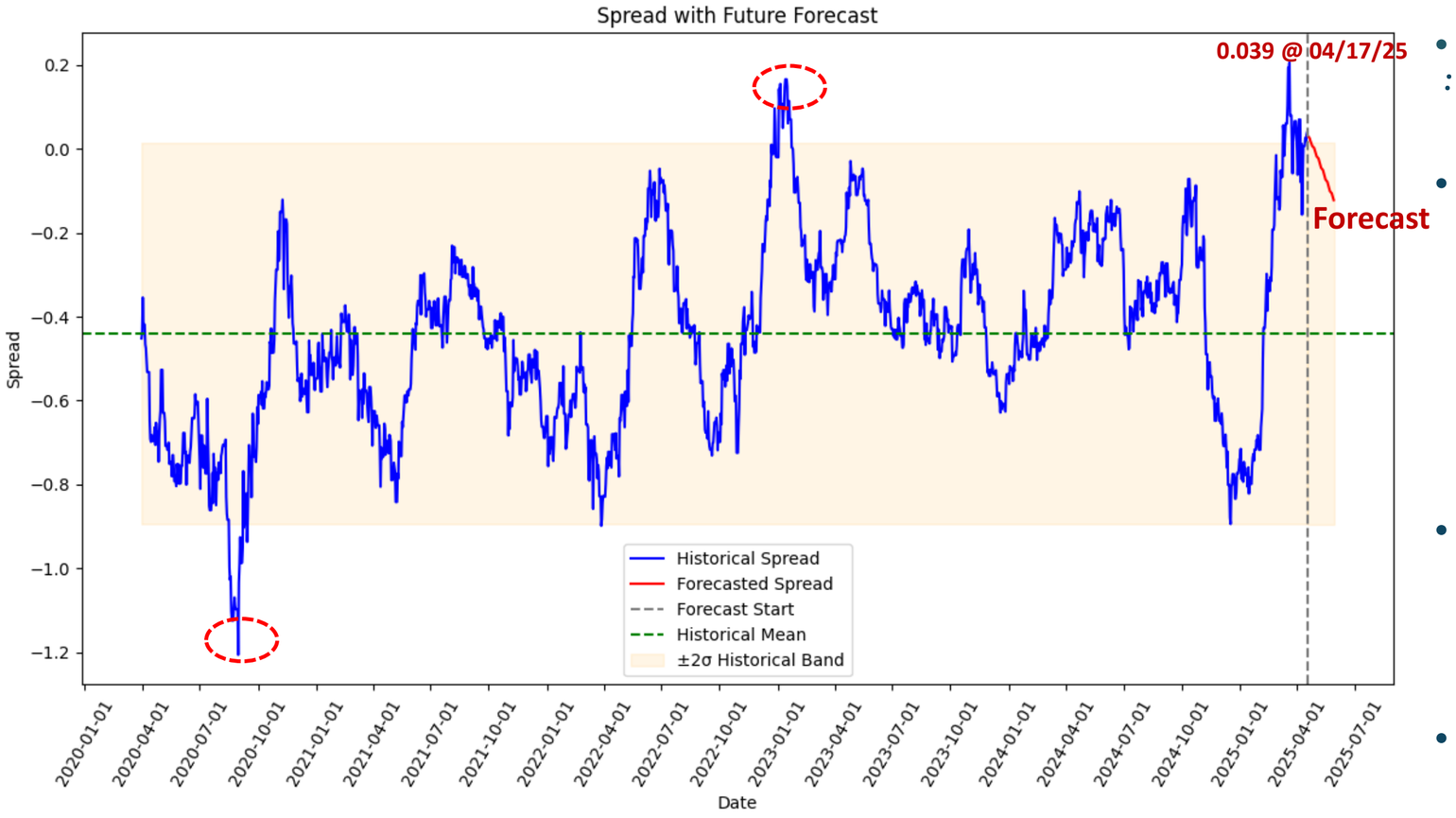
The Hurst Exponent $H$ measures the degree of long-term memory in a time series: $H < 0.5$: Mean-reverting series, $H = 0.5$: Random walk, $H > 0.5$: Trending series

The Hurst Exponent (0.426) is less than 0.5, indicating that the spread is mean-reverting. A value below 0.5 confirms that the spread has a tendency to revert to its mean, which is ideal for pairs trading.

Half-Life of Mean Reversion measures how quickly the spread reverts to its mean. It's derived from an OLS regression of the spread's first difference ($\Delta Z_t$) on its lagged value ($Z_{t-1}$): $\Delta Z_t = \lambda Z_{\{t-1\}} + \epsilon_t$, and $half - life = \frac{ln(2)}{\lambda}$

The half-life of mean reversion is the time it takes for the spread to revert halfway back to its mean after a deviation. A half-life of 51.6 days suggests that the mean reversion process is relatively slow, which may require a longer holding period for trades to become profitable.

# Quantitative Analysis (2) - *Long Tesla (Tsla.US) Short BYD (1211.HK)*




Note:

(1) The blue-line plot shows the normalized stock price spread between BYD and Tesla from March 31, 2020 to April 17, 2025 (1843 days), along with the historical mean and $\pm 2\sigma$ bands

(2) The red-line plot used ARIMA model to forecast 30-day downward trend of spread between BYD and Tesla starting from the current value (0.039) on April 17 and will reach around −0.1 by mid-May 2025.

(3) The forecast remains within the $\pm 2\sigma$ band, indicating no extreme deviations are expected in the next 30 days.

- $Spread = Ln(Price_{[BYD]}) - \beta * Ln(Price_{[Tesla]})$,
  : $\beta$ represnts hedge ratio of **0.897**
- The BYD-Tesla price spread exhibits clear mean-reverting behavior, as it repeatedly returns to the historical mean after deviations. e.g, after dipping to -1.2 in 2020 and peaking in 2023, it declined toward the mean finally.
- The spread's mean-reverting behavior and high percentage of time within the 2-SD band suggest that BYD-Tesla is good for pair stock trading.
- The spread stays within a normal range 96% of the time and is expected to drop from **0.039 on April 17, 2025** to the historical mean by mid-May 2025, indicating a potential **opportunity (sell BYD and buy Tesla)**

# Quantitative Analysis (3) - *Long Tesla (Tsla.US) Short BYD (1211.HK)*

**Spread Correlation Testing:**
**BYD-Tesal vs. HSI-NASDAQ COMP**

| Metric | Value |
|---|---|
| **Model Information** | |
| Dependent Variable | BYD-Tesla Return Spread |
| Independent Variable | HSI-NASDAQ Market Spread |
| Number of Observations | 1,843.0 |
| **Coefficients** | |
| Intercept (α) | 0.0008 |
| p-value | 0.2997 |
| Beta (β) | 1.4271 |
| p-value | 1.05e-167 |
| **Model Fit** | |
| R-squared | 0.3390 |

Note:

$$Spread\ (BYD\text{–}Tesla) = \alpha + \beta \times Spread(HSI\text{–}NASDAQ\ COMP) + \epsilon$$

(1) **Data Input:** Price and Index for 1843 days from March 31, 2020 to April 17, 2025

(2) **Coefficient Estimates:**

- **Intercept ($\alpha$): 0.0008**, the expected BYD-Tesla spread when the market spread is zero. **(p-value exceeds 0.05, not statistically significant)**
- **Beta ($\beta$)**: 1.4271, a 1% change in the HSI-NASDAQ market spread is associated with a 1.4271% change in the BYD-Tesla spread, suggesting a leveraged response. (p-value is extremely small, confirming high statistical significance)

(3) **Model Fit:**

- **R-squared: 0.3390**, 33.9% of the variance in the BYD-Tesla spread is explained by the HSI-NASDAQ spread.
- This moderate value reflects a partial but meaningful relationship, typical in financial return models where multiple factors influence outcomes.

- The intercept ($\alpha$) implies there is **NO** reliable evidence of a consistent outperformance of BYD over Tesla (or vice versa) independent of the market spread between Hang Seng Index (HSI) and NASDAP Composite Index (COMP) , reinforcing that the relationship of pair stock is primarily driven by the independent variable rather than a constant effect.
- The beta ($\beta$) coefficient suggests that the market spread is a key driver of the relative performance between BYD and Tesla, with Tesla's returns being more responsive to NASDAQ COMP's movements relative to BYD's sensitivity to HSI (Based on additional regression results).
- The R-squared of 0.3390 implies that while the HSI-NASDAQ market spread is a useful predictor for the BYD-Tesla price spread, it only explains about a third of the variation. The model provides a meaningful starting point for understanding relative performance but should not be relied upon exclusively.

# Trading Strategy and Risk Control - *Long Tesla (Tsla.US) Short BYD (1211.HK)*

## ☐ Pair Trading Strategy

- Entry timing: Adjust the long-short strategy dynamically as the price spread of pair stocks widens or narrows
- Exit timing: Close trading positions when the price spread reverts to its historical average
- Stop-loss setting: Clear the position if the spread widens beyond ±3 standard deviations or if the ADF test indicates a breakdown in cointegration

## ☐ Risk Control

- Model Risk： Rolling cointegration tests instead of static model backtesting.
  - ✓ Check the validation of assumption that the pair stocks' correlations will revert to historical norms.
  - ✓ Many pairs exhibits correlation breakdowns during market crises.
- HKD/USD Currency Risk: Mitigate HKD/USD exchange rate risk using forex futures or ETFs to hedge
- Trading Execution Risk: Hong Kong vs. NASDAQ market's trading hours may cause execution gaps. Divergent trading hours between the Hong Kong and NASDAQ markets create a 13-hour window where BYD and Tesla trade in isolation.
- Volatility Risk: Tesla's 90-day volatility is higher than BYD's, so it is necessary to adjust dollar amounts to balance risk.
- Liquidity asymmetry risk: The illiquidity measure shows BYD's Hong Kong shares have multiple higher costs than Tesla's NASDAQ-listed stock

*Thank you*